compile model

optimize = stochastic gradient descent

Loss = sparse categorical cross entropy

metric = accuracy.

Observation:

The loss decreases with each epoch, showing that the model is learning.

Accuracy improves steadily during training.

Testing: overall accuracy of the model on the entire dataset.

Result:

successfully build a simple feed forward neural network to recognize handwritten character.

22/8/25

Epoch	Loss
1	1.0547
2	0.3846
3	0.3246
4	0.2934
5	0.2689

Test Accuracy : 92.80%.

[1]:
```python
import numpy as np
import pandas as pd
```

[2]:
```python
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
```

[3]:
```python
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay, classi
from sklearn.linear_model import LogisticRegression
```

[4]:
```python
!pwd
```

```
/home/jupyter-ra2311047010010/DL
```

[9]:
```python
data = pd.read_csv('/home/jupyter-ra2311047010010/DL/breast_cancer_bd.csv')
```

[10]:
```python
###PREPROCESSING
```

[11]:
```python
data.head(15)
```

[11]:

	Sample code number	Clump Thickness	Uniformity of Cell Size	Uniformity of Cell Shape	Marginal Adhesion	Single Epithelial Cell Size	Bare Nuclei	Bland Chromatin	Normal Nucleoli	N
0	1000025	5	1	1	1	2	1	3	1	
1	1002945	5	4	4	5	7	10	3	2	
2	1015425	3	1	1	1	2	2	3	1	

```
[12]: data = data.drop('Sample code number', axis=1)
```

```
[13]: data = data.replace('?', np.nan)

      data['Bare Nuclei'] = pd.to_numeric(data['Bare Nuclei'])
      data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 699 entries, 0 to 698
Data columns (total 10 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   Clump Thickness              699 non-null    int64
 1   Uniformity of Cell Size      699 non-null    int64
 2   Uniformity of Cell Shape     699 non-null    int64
 3   Marginal Adhesion            699 non-null    int64
 4   Single Epithelial Cell Size  699 non-null    int64
 5   Bare Nuclei                  683 non-null    float64
 6   Bland Chromatin              699 non-null    int64
 7   Normal Nucleoli              699 non-null    int64
 8   Mitoses                      699 non-null    int64
 9   Class                        699 non-null    int64
dtypes: float64(1), int64(9)
memory usage: 54.7 KB
```

```
[14]: data['Bare Nuclei'] = data['Bare Nuclei'].fillna(data['Bare Nuclei'].median())
```

```
[15]: y = data['Class']
```

```
dtypes: float64(1), int64(9)
memory usage: 54.7 KB
```

[14]:
```
data['Bare Nuclei'] = data['Bare Nuclei'].fillna(data['Bare Nuclei'].median())
```

[15]:
```
y = data['Class']
X = data.drop('Class', axis=1)
```

[16]:
```
##SPLITTING AND TRAINING
```

[17]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size= 0.3)
```

[18]:
```
lr_model = LogisticRegression()

lr_model.fit(X_train, y_train)
```

[18]:

LogisticRegression

▼ Parameters

penalty	'l2'
dual	False
tol	0.0001
C	1.0
fit_intercept	True

l1_ratio	None

[19]:
```
##EVALUATION
```

[20]:
```
y_pred_lr = lr_model.predict(X_test)
score = accuracy_score(y_test,y_pred_lr)
print('Accuracy score: {}'.format(score))
```

```
Accuracy score: 0.9523809523809523
```

[21]:
```
cm = confusion_matrix(y_test, y_pred_lr)
print(cm)
```

```
[[134   6]
 [  4  66]]
```

[22]:
```
from sklearn.metrics import roc_auc_score
auc_score = roc_auc_score(y_test, y_pred_lr)
auc_score
```

[22]: 0.9500000000000001

[23]:
```
##VISUALIZING
```

[24]:
```
 import matplotlib.pyplot as plt
```

[25]:
```
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=lr_model.classes_)
```

lab3.ipynb (6) - JupyterLab | Breast Cancer Wisconsin (Diagn | Why use load_digits | Inbox (2,286) - rh0591@smist

Not secure 10.1.38.19/user/ra2311047010010/lab/tree/DL/lab3.ipynb

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Name	Last Modified
Untitled Fol...	next year
breast_canc...	next year
lab1.ipynb	next year
lab2.ipynb	next year
lab3.ipynb	next year

	code number	Thickness	of Cell Size	of Cell Shape	Adhesion	Epithelial Cell Size	Nuclei	Chromatin	Nucleoli
0	1000025	5	1	1	1	2	1	3	1
1	1002945	5	4	4	5	7	10	3	2
2	1015425	3	1	1	1	2	2	3	1
3	1016277	6	8	8	1	3	4	3	7
4	1017023	4	1	1	3	2	1	3	1
5	1017122	8	10	10	8	7	10	9	7
6	1018099	1	1	1	1	2	10	3	1
7	1018561	2	1	2	1	2	1	3	1
8	1033078	2	1	1	1	2	1	1	1
9	1033078	4	2	1	1	2	1	2	1
10	1035283	1	1	1	1	1	1	3	1
11	1036172	2	1	1	1	2	1	2	1
12	1041801	5	3	3	3	2	3	4	4
13	1043999	1	1	1	1	2	3	3	1
14	1044572	8	7	5	10	7	9	5	5

VARIABLES | CALLSTACK | BREAKPOINTS | SOURCE | KERNEL SOURCES

Simple 1 5 Python 3 (ipykernel) | Idle Mem: 1.02 GB Mode: Command Ln 3, Col 31 lab3.ipynb 1

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